

Reflection

In this conversation with ChatGPT, I asked it first to provide me with 5 MCQ questions. The questions weren't so hard, it asked about the importance of normalizing the features, it asked me to calculate the gradient and coefficient update step based on some numerical example it gave me, it also asked me about the effect of ridge regularization on the coefficients. I answered all the questions correctly even though for the numerical examples questions it told me my answers aren't correct but after it tried to show me the explanation it told me that my answers are actually correct.

After the MCQ questions, I asked it how to do normalization if the scale of the test data is not the same as the scale of the training data. Its answer was that normalization on the test data should be done using the statistical characteristics of the training data, which means that I should use the mean and variance of the training data not the test data.

Then I asked it to provide me with the mathematical equations and explanation for solving the linear regression problem analytically and I asked it to give me a code snippet to apply these equations on my dataset. The equations and the code are straight forward (simple linear algebra), however, for large datasets gradient descent is preferable and more stable according to ChatGPT.

After that I told it to explain to me what really makes collinearity in the features bad to the linear regression algorithm and how ridge regularization can help avoid this problem. The main idea is that collinear features make the model underdetermined and because of that there become infinitely many solutions for the coefficients, this results in no sensibility in the choice of the coefficients, it becomes more of a random way and therefore it affects generalization. Ridge regularization helps with that by forcing the solution with the smallest coefficients.

Then I asked it to provide me with a code that generates values for an OLS loss function that is convex but not strictly convex since I had hard time imagining how a loss function could be not strictly convex. However, I still couldn't understand the theory that ChatGPT used to generate these values of the non-strictly convex OLS loss function.

Finally, I asked it to give me extra 5 MCQs with a higher level of difficulty. The questions were related to the questions I asked during the conversation and they were relatively manageable.